



ENTRANCE EXAMINATION - XI

SWARAJYA MILITARY SCHOOL & ACADEMY

Khandi Pimpalgaon, Tq. Khultabad, Dist: Chh. Sambhajinagar

"Empowering Potential"

EXAM SESSION:	ENT : 2025	MAX. MARKS:	50 Marks
TIME ALLOWED:	1 Hour (60 Minutes)	CANDIDATE ROLL NO:	
CANDIDATE NAME:			

QUESTION BOOKLET – GENERAL GUIDELINES

- Strict Prohibition:** Do not carry any book, paper, mobile phone, or any other material inside the Examination Hall.
- Stationery:** Carry only permitted writing material with you in the Examination Hall.
- Result Notification:** The result of this examination will be communicated to your registered mobile number.
- Interview:** All successful candidates will be called for the personal interview round.
- Booklet Structure:** This Booklet contains **FIVE SECTIONS:** English, General Science, Mathematics, General Knowledge (Factual), and Essay Writing. Each MCQ section contains 10 Multiple Choice Questions (Total 40 MCQs, 40 Marks). Writing & language skills are assessed through Essay Writing (10 Marks).
- Selecting Answers:** Each question has four responses (A, B, C, & D). Select the response which you consider the best and mark it on the Answer Sheet. In any case, choose only one response for each question.
- Marking on OMR Sheet:** Indicate your choice by completely blackening the respective circle. Do NOT merely cross or tick the circle.
- Single Marking Rule:** Indicate your correct answer by blackening **ONLY ONE** circle per question. Any answer having more than one blackened circle will be treated as an incorrect answer.
- Rough Work:** All answers must be marked on the Answer Sheet only (OMR Sheet). Do not write anything on this Test Booklet. Separate rough work papers will be provided upon request.
- Booklet Integrity:** Check immediately upon commencement of the examination that this test booklet does not have any unprinted, torn, or missing pages. If so, get it replaced by a new one immediately.
- Document Submission:** Return the Test Booklet, rough work papers, and the Answer Sheet immediately when asked to do so. You are not permitted to take away any paper concerning the examination.
- Verification:** Before starting, ensure that you have filled in all the complete information at the top of the Answer Sheet. Incomplete Answer Sheets will be regarded as invalid.

Read the passage and answer questions 1 to 5:

Necessity, they say, is the mother of invention. And Indian doctors have been quite creative when hamstrung by few or no tools to perform specific surgeries. They simply design it themselves at one-fourth the price they are sold abroad. In fact, some of their innovations are priced at as much as hundreds of dollars abroad.

Take 47-year-old Dr Burjor P Banaji, pioneer of Lasik surgery in India. He's invented over a dozen surgical instruments. When this senior eye surgeon at Max Eye Care started Lasik, there were few surgeons doing it worldwide and no specific instruments were available either. "As I want things super-perfect, I designed a whole slew of instruments that made my surgery more efficient," says Banaji. The most popular instruments are Banaji Lasik Shield and Banaji Lasik Spatula and Cannula. "It was simple. I had the designs in my head. Putting them down on paper was the simplest thing," he says. Instruments manufacturers and large multinationals in the US snapped them up. "They would send me computer generated drawings which I would correct and send back. Their level of execution was astounding. Within two weeks of the designs being finalized, the instruments were in the world market."

His instruments are priced at hundreds of dollars each in the US, and are also sold in Switzerland, South America, Korea, Eastern Europe, Africa and Japan. They're available in India at a fraction of the price.

1. Why have some Indian doctors created their own surgical tools?

- a) they have no tools to perform specific surgeries b) they have a hamstring problem
c) the tools they get from abroad cost four times as much d) they can sell these tools at a very high price

2. What has Dr Burjor P Banaji created?

- a) Lasik surgery b) Max Eye Care Centre
c) Two surgical instruments called Banaji Shield & Spatula d) More than a dozen instruments for operating on the eye

3. Where does Dr. Banaji get the instruments manufactured?

- a) India b) The United States c) Switzerland d) Japan

4. In the context of the passage, the term "hamstrung" refers to:

- a) restricted b) helped c) harmed d) liberated

5. What does the phrase "slew of instruments" refer to?

- a) a wide range of instruments b) instruments used for slaying
c) tools of a similar nature d) surgical instruments

6. Choose the correct antonym of the word: Transmit

- a) Televisе b) Withhold c) Reply d) Show

7. Choose the correct antonym of the word: Revealed

- a) Naked b) Concealed c) Open d) Bare

8. That good boy always speaks _____ :

- a) Polite b) Politely c) Impolite d) Politeness

9. Change the noun into verb form: Intention

- a) intension b) intend c) intensity d) intentional

10. 'Shoes' is related to 'Leather' in the same way as 'Rubber' is related to:

- a) Plastic b) Polythene c) Latex d) Chappal

11. Two electric bulbs have resistances in the ratio 1 : 2. If they are connected in series, the electrical energy consumed by them will be in the ratio:

- (a) 1:2 (b) 2:1 (c) 4:1 (d) 1:1

12. Group _____ in the modern periodic table contains the alkaline earth metals.

- (a) one (b) two (c) zero (d) seventeen

13. When two or more chemical substances combine to form a single product, the reaction is known as a:

- (a) Combination reaction (b) Decomposition reaction
(c) Displacement reaction (d) Double displacement reaction

14. The pH scale helps in measuring the concentration of _____ ions in solutions.

- (a) hydrogen (b) nitrogen (c) oxygen (d) carbon

15. _____ is the only non-metallic element which is liquid at standard room temperature.

- (a) Bromine (b) Sulphur (c) Carbon (d) Mercury

16. Which hydrocarbon gas is commonly known as "marsh gas"?

- (a) Methane (b) Ethane (c) Propane (d) Ethene

17. Which of the following is a large muscular blood vessel that carries oxygenated blood away from the heart?

- (a) Artery (b) Vein (c) Capillary (d) Nerve

18. Which of the following vitamins is NOT a member of the vitamin B complex?

- (a) Thiamine (b) Riboflavin (c) Folic acid (d) Ascorbic acid

19. Fungi are heterotrophic organisms that lack which photosynthetic pigment:

- (a) Oxygen (b) Carbon dioxide (c) Chlorophyll (d) Cell wall

20. Which fundamental physiological characteristic defines a reptile?

- (a) Cold blooded (b) Warm blooded (c) Non-hearing (d) Feathers

21. In an Arithmetic Progression (A.P.), if first term $a = 3$ and common difference $d = 5$, what is the value of t_{11} ?

- a) 53 b) 58 c) 85 d) 35

22. What is the value of k , if one root of the quadratic equation $kx^2 - 7x + 12 = 0$ is 3?

- a) 1 b) -1 c) 3 d) -3

23. For drawing the linear graph of $2x + 5y = 16$, if $x = 3$, what is the corresponding value of y ?

- a) 2 b) -2 c) $1/2$ d) $-1/2$

24. Three fair coins are tossed simultaneously. If P is the event of getting at least two heads, then P is equal to:

- a) { HHH, HHT, HTH, THH } b) { HHT, HTH, THH }
c) { HHT } d) { HTH, THH }

25. In right triangle ΔABC , $\angle B = 90^\circ$, $\angle A = 60^\circ$. If hypotenuse $AC = 24$ cm, what is the length of side AB ?

- a) 12 cm b) $12\sqrt{3}$ cm c) 16 cm d) $16\sqrt{3}$ cm

26. If $\Delta ABC \sim \Delta PQR$, with $AB = 6$ cm, $BC = 4$ cm, $AC = 8$ cm, and $PQ = 9$ cm, what is the perimeter of ΔPQR ?

- a) 54 cm b) 27 cm c) 18 cm d) 36 cm

27. What is the point of concurrence of the internal angle bisectors of a triangle known as?

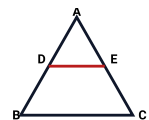
- a) Circumcentre b) Incentre c) Orthocentre d) Centroid

28. What is the curved surface area of a right circular cone having base diameter 14 cm and vertical height 24 cm? (Take $\pi = 22/7$)

- a) 110 cm^2 b) 220 cm^2 c) 440 cm^2 d) 550 cm^2

29. In the given figure, $DE \parallel BC$. If $AB = 7.6$ cm and $AD = 1.9$ cm, then the ratio $AE : EC$ is:

- a) 1:3 b) 1:4 c) 4:1 d) 3:1



30. What is the nature of the roots of the quadratic equation $4x^2 - 8x + 9 = 0$?

- a) Real and equal b) Not real (Complex)
c) Real and unequal d) Purely imaginary only

31. Which of the following statements is NOT true?

- (a) Nirbhay is a missile of 1000 km
(b) BrahMos missile is developed by DRDO
(c) Astra is an Air-to-Surface missile
(d) The range of Agni-III is 3000 to 5000 km

32. Dhruv is an Indian:

- (a) Main battle tank
(b) Advanced Light Helicopter (ALH)
(c) Multi-barrel rocket system
(d) Surface-to-air missile

33. Which country is historically one of the biggest importers of arms and ammunition in the world?

- (a) Iran
(b) Afghanistan
(c) Pakistan
(d) India

34. From which foreign partner country does India procure a significant proportion of its defence equipment?

- (a) Russia
(b) USA
(c) Israel
(d) China

35. What is NOT true regarding the BrahMos supersonic cruise missile?

- (a) Developed by DRDO with help of the US
(b) Supersonic missile launched from land, air & sea
(c) Original base range of 290 km
(d) Can fly as low as 10 meters altitude

36. Where is the headquarters of Mazagon Dock Shipbuilders Limited located?

- (a) Visakhapatnam
(b) Kolkata
(c) Mumbai
(d) Kandla

37. On which date is Indian Army Day observed annually?

- (a) 4 December
(b) 15 January
(c) 28 February
(d) 1 December

38. Which of the following is the oldest paramilitary force in India?

- (a) CRPF
(b) BSF
(c) ITBP
(d) Assam Rifles

39. Which item belongs to a different weapon class than the others (is NOT a battle tank)?

- (a) Vijayanta
(b) T-55
(c) Arjun
(d) Agni

40. Bhabha Atomic Research Centre (BARC) is situated in which Indian state?

- (a) Karnataka
(b) Maharashtra
(c) Uttarakhand
(d) Delhi

PART V – ESSAY WRITING

10 Marks

CHOOSE AND WRITE AN ESSAY ON ANY ONE OF THE FOLLOWING TOPICS (100–150 WORDS):

Option 1: The Effect of Pollution on the Environment

– OR –

Option 2: The Impact of Technology on Education

Instructions: Structure your essay logically with an Introduction, Main Body Paragraphs, and a Conclusion. Write legibly on your Answer Sheet.